

UNIVERSITÉ HASSAN II – CASABLANCA

FACULTÉ DES SCIENCES BEN M'SICK
DÉPARTEMENT DE MATHÉMATIQUES ET
INFORMATIQUE

Master Data Science & Big Data

Module MLOps & DataOps

TP Chapitre 2

*Pipelines agiles, versionnement et orchestration avec dlt,
dbt et Dagster*

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I. Captures d'écran des commandes importantes

1. Installation les dépendances

```
Activities Terminal 31 May 13:53 med@vm-mohamed: ~/tp_chapitre2_pipeline

Downloading sqllight-30.8.0-py3-none-any.whl (687 kB)
287/1027.0 kB 1.0 MB/s 0:00:00
Downloading starlette-1.2.1-py3-none-any.whl (73 kB)
Downloading tenacity-9.1.4-py3-none-any.whl (28 kB)
Downloading text_unidecode-1.3-py2.py3-none-any.whl (78 kB)
Downloading tonkit-0.15.0-py3-none-any.whl (41 kB)
Downloading toposort-1.10-py3-none-any.whl (8.5 kB)
Downloading typing_inspection-0.4.2-py3-none-any.whl (14 kB)
Downloading tzdata-2026.2-py2.py3-none-any.whl (349 kB)
Downloading tzpp-4.1.0-py3-none-any.whl (10 kB)
Downloading docstring_parser-0.18.0-py3-none-any.whl (22 kB)
Downloading filelock-3.29.0-py3-none-any.whl (39 kB)
Downloading nako-1.3.12-py3-none-any.whl (78 kB)
Downloading platformdirs-4.10.0-py3-none-any.whl (22 kB)
Downloading ply-3.11-py2.py3-none-any.whl (49 kB)
Downloading python_dotenv-1.2.2-py3-none-any.whl (22 kB)
Downloading structlog-25.5.0-py3-none-any.whl (72 kB)
Downloading tabulate-0.10.0-py3-none-any.whl (29 kB)
Downloading universal_pathlib-0.3.10-py3-none-any.whl (83 kB)
Downloading pathlib_abc-0.5.2-py3-none-any.whl (19 kB)
Downloading uvicorn-0.48.0-py3-none-any.whl (71 kB)
Downloading h11-0.16.0-py3-none-any.whl (37 kB)
Downloading httpx-0.8.0-cp310-cp310-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (450 kB)
Downloading uvloop-0.22.1-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (3.7 MB)
272/778.7 MB 1.0 MB/s 0:00:00
Downloading watchfiles-1.2.0-cp310-cp310-manylinux1_x86_64.manylinux_2_17_x86_64.manylinux2014_x86_64.whl (457 kB)
Downloading websockets-16.0-cp310-cp310-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (183 kB)
Installing collected packages: toposort, text-unidecode, pytz, pytimeparse, ply, parsedatetime, leather, daff, antir4-python3-runtime, ripp, websockets, watchdog, uvloop, urllib3, tzdata, typing-extensions, tqdm, tonkit, tenacity, tabulate, sqlparse, sqllight, smmap, six, simplejson, setuptools, server, rps-py, pyyaml, python-slugify, python-dotenv, pygments, protobuf, propcache, pluggy, platformdirs, pathvalid, ate, pathspec, pathlib-abc, packaging, orjson, orderly-set, numpy, networkx, msgpack, more-iter-tools, mdurl, MarkupSafe, jsonpath-ng, isodate, idna, humanize, humanfriendly, httpx, h11, greenlet, graphql-core, giturlparse, fsspec, filelock, duckdb, docstring-parser, dbt-extractor, dagster-pipes, colorama, click, charset-normalizer, certifi, backoff, Babel, attrs, annotated-types, uvicorn, universal-pathlib, typing-inspection, structlog, sqllight, requirements-parser, requests, referencing, python-dateutil, pydantic-core, multidict, mashumaro, markdown-it-py, Mako, Jinja2, importlib-metadata, grpcio, graphql-relay, gitdb, exceptiongroup, deepdiff, dbt-protos, coloredlogs, agate, yarl, snowlow-tracker, rich, requests-toolbelt, pydantic, pendulum, pandas, jsonschema-specifications, grpcio-health-checking, graphene, gitypython, an-yio, alembic, watchfiles, starlette, rich-argparse, jsonschema, gql, dagster-shared, dlt, dbt-semantic-interfaces, dbt-common, dagster, dbt-adapters, dagster-graphql, dbt-core, dagster-webserver, dbt-duckdb
Attempting uninstall: setuptools
Found existing installation: setuptools 59.6.0
Uninstalling setuptools-59.6.0:
Successfully uninstalled setuptools-59.6.0
Successfully installed Babel-2.18.0 Mako-1.3.12 MarkupSafe-3.0.3 agate-1.9.1 alembic-1.18.4 annotated-types-0.7.0 antir4-python3-runtime-4.13.2 anyio-4.13.0 attrs-26.1.0 backoff-2.2.1 certifi-2026.5.20 charset-normalizer-3.4.7 click-8.1.1 colorama-0.4.6 coloredlogs-14.0 daff-1.4.2 dagster-1.13.7 dagster-graphql-1.13.7 dagster-pipes-1.13.7 dagster-shared-1.13.7 dagster-webserver-1.13.7 dbt-adapters-1.24.2 dbt-common-1.3.8.0 dbt-core-1.11.11 dbt-duckdb-1.10.1 dbt-extractor-0.6.0 dbt-protos-1.0.514 dbt-semantic-interfaces-0.9.0 deepdiff-8.6.2 dlt-1.27.2 docstring-parser-0.18.0 duckdb-1.5.3 exceptiongroup-1.3.1 filelock-3.29.0 fsspec-2026.4.0 gitdb-4.0.12 gitypython-3.1.50 giturlparse-0.14.0 gql-4.0.0 graphene-3.4.3 graphql-core-3.2.0 graphql-relay-3.2.0 greenlet-3.5.1 grpcio-1.80.0 grpcio-health-checking-1.80.0 h11-0.16.0 httpx-0.8.0 humanfriendly-10.0 humanize-4.15.0 idna-3.17 importlib-metadata-8.0 isodate-0.7.2 Jinja2-3.1.0 jsonpath-ng-1.7.0 jsonschema-4.20.0 jsonschema-specifications-2025.9.1 leather-0.4.1 markdown-it-py-4.2.0 mashumaro-3.14 mdurl-0.1.2 more-iter-tools-10.8.0 msgpack-1.1.2 multidict-6.7.1 networkx-3.4.2 numpy-2.2.6 orderly-set-5.5.0 orjson-3.11.9 packaging-26.2 pandas-2.3.3 parsedatetime-2.6 pathlib-abc-0.5.2 pathspec-0.12.1 pathvaldate-3.3.1 pendulum-3.2.0 platformdirs-4.10.0 pluggy-1.6.0 ply-3.11 propcache-0.5.2 protobuf-6.33.6 pydantic-2.13.4 pydantic-core-2.46.4 pygments-2.20.0 python-dateutil-2.9.0.post0 python-dotenv-1.2.2 pythonslugify-8.0.4 pytimeparse-1.1.8 pytz-2026.2 pyyaml-6.0.3 referencing-0.37.0 requests-2.34.2 requests-toolbelt-1.0.0 requirements-parser-0.13.0 rich-15.0.0 rich-argparse-1.8.0 rps-py-0.30.0 server-3.0.4 setuptools-68.0.1 simplejson-3.11.0 six-1.17.0 smmap-5.0.3 snowlow-tracker-1.10 sqlalchemypy-2.0.50 sqllight-30.8.0 sqlparse-0.5.5 starlette-1.2.1 structlog-25.5.0 tabulate-0.10.0 tenacity-9.1.4 text-unidecode-1.3 toml-2.4.3 tonkit-0.15.0 toposort-1.10 tqdm-4.67.3 typing-inspection-0.4.2 typing-inspection-4.15.0 typing-inspection-0.4.2 tzdata-2026.2 universal-pathlib-0.3.10 urllib3-2.7.0 uvicorn-0.48.0 uvloop-0.22.1 watchdog-6.0.0 watchfiles-1.2.0 websockets-16.0 yarl-1.24.2 tzpp-4.1.0
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ pip freeze > requirements.txt
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$
```

2. Initialisation Git

```
Activities Terminal 31 May 14:12 med@vm-mohamed: ~/tp_chapitre2_pipeline

(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ git init
Initialized empty Git repository in /home/med/tp_chapitre2_pipeline/.git/
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ git config --global init.defaultBranch main
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ cat > .gitignore << "cor"
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$
__pycache__/
target/
dbt_packages/
*.duckdb
EOF
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ git add
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ git commit -m "Initialisation du projet pipeline"
[main (root-commit) f3b8d8f] Initialisation du projet pipeline
Committer: med <med@vm-mohamed.muguest.virtuallab.org>
Your name and e-mail address were configured automatically, based on your username and hostname. Please check that they are accurate.
You can suppress this message by setting then explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

2 files changed, 129 insertions(+)
create mode 100644 .gitignore
create mode 100644 requirements.txt
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$
```

3. Création de l'arborescence et de fichier source ; Étape 1 — Ingestion

```
Activities Terminal 31 May 14:47 med@vm-mohamed: ~/tp_chapitre2_pipeline

(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ git config --global init.defaultBranch main
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ cat > .gitignore << 'EOF'
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$
target/
__pycache__
dbt_packages/
*.duckdb
EOF
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ git add .
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ git commit -m "Initialisation du projet pipeline"
[main (root-commit) f3b0dd8f] Initialisation du projet pipeline
 1 file changed, 1 insertion(+)
 commit: med <med@vm-mohamed.myquest.virtualbox.org>
Your name and e-mail address were configured automatically, based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

2 files changed, 129 insertions(+), 1 deletion(-)
 create mode 100644 .gitignore
 create mode 100644 requirements.txt
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ mkdir -p data pipeline dbt_pipeline/models .github/workflows
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ touch data/ventes.csv
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ gedit data/ventes.csv
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ touch pipeline/ingest.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ gedit pipeline/ingest.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ python pipeline/ingest.py
Ingestion terminée : ventes_raw créée dans DuckDB
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ git add pipeline/ingest.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ git commit -m "Ajout de l'étape d'ingestion"
[main 079692a] Ajout de l'étape d'ingestion
 1 file changed, 10 insertions(+), 1 deletion(-)
 commit: med <med@vm-mohamed.myquest.virtualbox.org>
Your name and e-mail address were configured automatically, based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 10 insertions(+), 1 deletion(-)
 create mode 100644 pipeline/ingest.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$
```

4. Étape 2 — Validation minimale

```
Activities Terminal 31 May 14:52 med@vm-mohamed: ~/tp_chapitre2_pipeline

git commit --amend --reset-author

2 files changed, 129 insertions(+), 1 deletion(-)
 create mode 100644 .gitignore
 create mode 100644 requirements.txt
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ mkdir -p data pipeline dbt_pipeline/models .github/workflows
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ touch data/ventes.csv
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ gedit data/ventes.csv
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ touch pipeline/ingest.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ gedit pipeline/ingest.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ python pipeline/ingest.py
Ingestion terminée : ventes_raw créée dans DuckDB
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ git add pipeline/ingest.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ git commit -m "Ajout de l'étape d'ingestion"
[main 079692a] Ajout de l'étape d'ingestion
 1 file changed, 10 insertions(+), 1 deletion(-)
 commit: med <med@vm-mohamed.myquest.virtualbox.org>
Your name and e-mail address were configured automatically, based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 10 insertions(+), 1 deletion(-)
 create mode 100644 pipeline/ingest.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ touch pipeline/validate.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ gedit pipeline/validate.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ python pipeline/validate.py
Validation réussie : schéma et qualité minimale OK
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ git add pipeline/validate.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$ git commit -m "Ajout de l'étape de validation minimale"
[main f4c072] Ajout de l'étape de validation minimale
 1 file changed, 22 insertions(+), 1 deletion(-)
 commit: med <med@vm-mohamed.myquest.virtualbox.org>
Your name and e-mail address were configured automatically, based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 22 insertions(+), 1 deletion(-)
 create mode 100644 pipeline/validate.py
(.venv) med@vm-mohamed:~/tp_chapitre2_pipeline$
```

5. Étape 3 — Transformation avec dbt

```
Activities Terminal 31 May 14:57 med@vm-mohamed: ~/tp_chapitre2_pipeline/dbt_pipeline

after doing this, you may fix the identity used for this commit with:

git commit --amend --reset-author

1 file changed, 22 insertions(+)
create mode 100644 pipeline/validate.py
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ touch dbt_pipeline/dbt_project.yml
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ gedit dbt_pipeline/dbt_project.yml
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ touch dbt_pipeline/profiles.yml
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ gedit dbt_pipeline/profiles.yml
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ touch dbt_pipeline/models/ventes_clean.sql
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ gedit dbt_pipeline/models/ventes_clean.sql
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ touch dbt_pipeline/models/ventes_resume.sql
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ gedit dbt_pipeline/models/ventes_resume.sql
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ cd dbt_pipeline
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline/dbt_pipeline$ dbt debug --profiles-dir .
13:56:53 Running with dbt=1.11.11
13:56:53 dbt version: 1.11.11
13:56:53 python version: 2.10.12
13:56:53 python path: /home/med/tp_chapitre2_pipeline/.venv/bin/python
13:56:53 os info: Linux-6.8.0-106-generic-x86_64-with-glibc2.35
13:56:53 Using profiles dir at .
13:56:53 Using profiles.yml file at ./profiles.yml
13:56:53 Using dbt_project.yml file at /home/med/tp_chapitre2_pipeline/dbt_pipeline/dbt_project.yml
13:56:53 adapter type: duckdb
13:56:53 adapter version: 1.10.1
13:56:53 Configuration:
13:56:53   profiles.yml file [OK found and valid]
13:56:53   dbt_project.yml file [OK found and valid]
13:56:53   Required dependencies:
13:56:53     - git [OK found]
13:56:53 Connection:
13:56:53   database: ventes
13:56:53   schema: main
13:56:53   path: ../ventes.duckdb
13:56:53   config_options: None
13:56:53   extensions: None
13:56:53   settings: {}
13:56:53   external_root: .
13:56:53   use_credential_provider: None
13:56:53   attach: None
13:56:53   filesystems: None
13:56:53   remote: None
13:56:53   plugins: None
13:56:53   disable_transactions: False
13:56:53   Registered adapter: duckdb1.10.1
13:56:54 Connection test: [OK connection ok]
13:56:54 All checks passed!
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline/dbt_pipeline$
```

```
Activities Terminal 31 May 14:57 med@vm-mohamed: ~/tp_chapitre2_pipeline

13:56:53 attach: None
13:56:53 filesystems: None
13:56:53 remote: None
13:56:53 plugins: None
13:56:53 disable_transactions: False
13:56:53 Registered adapter: duckdb1.10.1
13:56:54 Connection test: [OK connection ok]
13:56:54 All checks passed!
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline/dbt_pipeline$ dbt run --profiles-dir .
13:57:16 Running with dbt=1.11.11
13:57:16 Registered adapter: duckdb1.10.1
13:57:16 Unable to do partial parsing because saved manifest not found. Starting full parse.
13:57:18 Found 2 models, 485 macros
13:57:18 Concurrency: 1 threads (target='dev')
13:57:18
13:57:18 1 of 2 START sql table model main.ventes_clean ..... [RUN]
13:57:18 1 of 2 OK created sql table model main.ventes_clean ..... [OK in 0.17s]
13:57:18 2 of 2 START sql table model main.ventes_resume ..... [RUN]
13:57:18 2 of 2 OK created sql table model main.ventes_resume ..... [OK in 0.05s]
13:57:18 Finished running 2 table models in 0 hours 0 minutes and 0.41 seconds (0.41s).
13:57:18 Completed successfully
13:57:18
13:57:18 Done. PASS=2 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=2
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline/dbt_pipeline$ cd ..
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ git add dbt_pipeline
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ git commit -m "Ajout des modèles dbt"
[main 2eb0443] Ajout des modèles dbt
3 files changed, 321 insertions(+)
create mode 100644 dbt_pipeline/user.yml
create mode 100644 dbt_pipeline/dbt_project.yml
create mode 100644 dbt_pipeline/logs/dbt.log
create mode 100644 dbt_pipeline/models/ventes_clean.sql
create mode 100644 dbt_pipeline/models/ventes_resume.sql
create mode 100644 dbt_pipeline/profiles.yml
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$

After doing this, you may fix the identity used for this commit with:

git commit --amend --reset-author

6 files changed, 321 insertions(+)
create mode 100644 dbt_pipeline/user.yml
create mode 100644 dbt_pipeline/dbt_project.yml
create mode 100644 dbt_pipeline/logs/dbt.log
create mode 100644 dbt_pipeline/models/ventes_clean.sql
create mode 100644 dbt_pipeline/models/ventes_resume.sql
create mode 100644 dbt_pipeline/profiles.yml
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$
```

6. Étape 4 — Test simple avec dbt

```
git commit --amend --reset-author

6 files changed, 221 insertions(+)
create mode 100644 dbt_pipeline/.user.yml
create mode 100644 dbt_pipeline/dbt_project.yml
create mode 100644 dbt_pipeline/logs/dbt.log
create mode 100644 dbt_pipeline/models/ventes_clean.sql
create mode 100644 dbt_pipeline/models/ventes_resume.sql
create mode 100644 dbt_pipeline/profiles.yml
(.venv) nedgvm-mohamed:/tp_chapitre2_pipeline$ touch dbt_pipeline/models/schema.yml
(.venv) nedgvm-mohamed:/tp_chapitre2_pipeline$ git add dbt_pipeline/models/schema.yml
(.venv) nedgvm-mohamed:/tp_chapitre2_pipeline$ cd dbt_pipeline
(.venv) nedgvm-mohamed:/tp_chapitre2_pipeline$ dbt test --profiles-dir .
14:01:45 Running with dbt=1.11.11
14:01:45 Registered adapter: duckdb=1.10.1
14:01:46 Found 2 models, 3 data tests, 485 macros
14:01:46 Concurrency: 1 threads (target='dev')
14:01:46
14:01:46 1 of 3 START test not_null_ventes_clean_produit ..... [RUN]
14:01:46 1 of 3 PASS not_null_ventes_clean_produit ..... [PASS in 0.05s]
14:01:46 2 of 3 START test not_null_ventes_clean_quantile ..... [RUN]
14:01:46 2 of 3 PASS not_null_ventes_clean_quantile ..... [PASS in 0.02s]
14:01:46 3 of 3 START test not_null_ventes_resume_categorie ..... [RUN]
14:01:46 3 of 3 PASS not_null_ventes_resume_categorie ..... [PASS in 0.03s]
14:01:46
14:01:46 Finished running 3 data tests in 0 hours 0 minutes and 0.21 seconds (0.21s).
14:01:46
14:01:46 Completed successfully
14:01:46
14:01:46 Done. PASS=3 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=3
(.venv) nedgvm-mohamed:/tp_chapitre2_pipeline$ git add dbt_pipeline/models/schema.yml
(.venv) nedgvm-mohamed:/tp_chapitre2_pipeline$ git commit -m "Ajout des tests simples dbt"
[main b02d96c] Ajout des tests simples dbt
 1 file changed, 17 insertions(+)
create mode 100644 dbt_pipeline/models/schema.yml
(.venv) nedgvm-mohamed:/tp_chapitre2_pipeline$
```

7. Étape 5 — Orchestration avec Dagster

```
(.venv) nedgvm-mohamed:/tp_chapitre2_pipeline$ dagster dev -f pipeline/orchestrate.py
/home/nedgvm-mohamed/.venv/lib/python3.10/site-packages/click/core.py:1077: SupersessionWarning: Function 'dev_command' is superseded and its usage is discouraged. Use 'dg dev' instead.
  return callback(*args, **kwargs)
2026-05-31 15:17:46 +0100 - dagster - INFO - Using temporary directory /home/nedgvm-mohamed/.tmp_dagster_home_2hapydm7 for storage. This will be removed when dagster dev exits.
2026-05-31 15:17:46 +0100 - dagster - INFO - To persist information across sessions, set the environment variable DAGSTER_HOME to a directory to use.
2026-05-31 15:17:47 +0100 - dagster - INFO - Launching Dagster services...
2026-05-31 15:17:58 +0100 - dagster-daemon - INFO - Instance is configured with the following daemons: ['AssetDaemon', 'BackfillDaemon', 'FreshnessDaemon', 'QueuedRunCoordinatorDaemon', 'SchedulerDaemon', 'SensorDaemon']
2026-05-31 15:17:51 +0100 - dagster-webserver - INFO - Serving dagster-webserver on http://127.0.0.1:3080 in process 20986
2026-05-31 15:25:35 +0100 - dagster-daemon - INFO - Received interrupt, shutting down daemon threads...
2026-05-31 15:25:35 +0100 - dagster - INFO - KeyboardInterrupt received
2026-05-31 15:25:35 +0100 - dagster - INFO - Shutting down Dagster services...
2026-05-31 15:25:35 +0100 - dagster-daemon - INFO - Daemon threads shut down.
2026-05-31 15:25:35 +0100 - dagster-webserver - INFO - Server for dagster-webserver was shut down.
2026-05-31 15:25:36 +0100 - dagster - INFO - Dagster services shut down.
(.venv) nedgvm-mohamed:/tp_chapitre2_pipeline$ git add pipeline/orchestrate.py
(.venv) nedgvm-mohamed:/tp_chapitre2_pipeline$ git commit -m "Ajout de l'orchestration Dagster"
[main ae7cb2c] Ajout de l'orchestration Dagster
 1 file changed, 32 insertions(+)
create mode 100644 pipeline/orchestrate.py
(.venv) nedgvm-mohamed:/tp_chapitre2_pipeline$
```

8. Étape 6 — CI simple

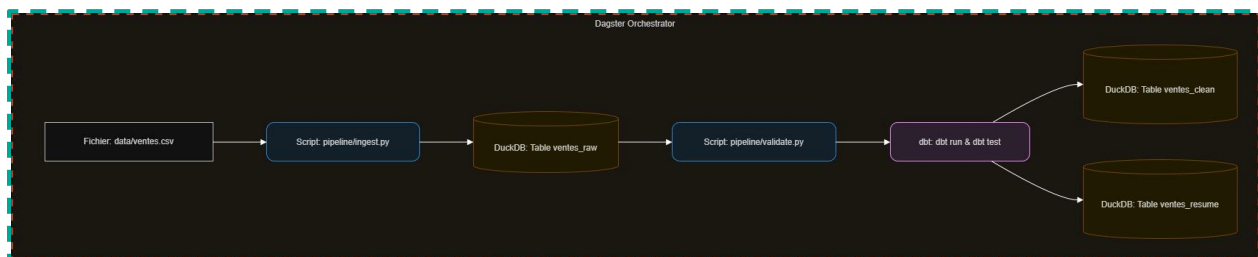
```
Activities Terminal 31 May 15:29 med@vm-mohamed: ~/tp_chapitre2_pipeline

/home/med/tp_chapitre2_pipeline/.venv/lib/python3.10/site-packages/click/core.py:877: SupersessionWarning: Function 'dev_command' is superseded and its usage is discouraged. Use 'dg dev' instead.
  return callback(*args, **kwargs)
2026-05-31 15:17:46 +0100 - dagster - INFO - Using temporary directory /home/med/tp_chapitre2_pipeline/.tmp_dagster_home_2happydn7 for storage. This will be removed when dagster dev exits.
2026-05-31 15:17:46 +0100 - dagster - INFO - To persist information across sessions, set the environment variable DAGSTER_HOME to a directory to use.
2026-05-31 15:17:47 +0100 - dagster - INFO - Launching Dagster services...
2026-05-31 15:17:50 +0100 - dagster.daemon - INFO - Instance is configured with the following daemons: ['AssetDaemon', 'BackfillDaemon', 'FreshnessDaemon', 'QueuedRunCoordinatorDaemon', 'SchedulerDaemon', 'SensorDaemon']
2026-05-31 15:17:51 +0100 - dagster-webserver - INFO - Serving dagster-webserver on http://127.0.0.1:3000 in process 20986
^C2026-05-31 15:25:35 +0100 - dagster.daemon - INFO - Received interrupt, shutting down daemon threads...
2026-05-31 15:25:35 +0100 - dagster - INFO - KeyboardInterrupt received
2026-05-31 15:25:35 +0100 - dagster - INFO - Shutting down Dagster services...
2026-05-31 15:25:35 +0100 - dagster.daemon - INFO - Daemon threads shut down.
2026-05-31 15:25:35 +0100 - dagster-webserver - INFO - Server for dagster-webserver was shut down.
2026-05-31 15:25:36 +0100 - dagster - INFO - Dagster services shut down.
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ git add pipeline/orchestrate.py
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ git commit -m "Ajout de l'orchestration Dagster"
[main ae7cb2c] Ajout de l'orchestration Dagster
 1 file changed, 22 insertions(+)
 create mode 100644 pipeline/orchestrate.py
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ touch .github/workflows/ci.yml
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ gedit .github/workflows/ci.yml
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ git add .github/workflows/ci.yml
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$ git commit -m "Ajout d'une CI simple"
[main 939f576] Ajout d'une CI simple
 1 file changed, 29 insertions(+)
 create mode 100644 .github/workflows/ci.yml
(.venv) med@vm-mohamed: ~/tp_chapitre2_pipeline$
```


II. Historique Git

```
med@vm-mohamed: ~/tp_chapitre2_pipeline
git log --oneline
939f576 (HEAD -> main) Ajout d'une CI simple
ae7cbc2 Ajout de l'orchestration Dagster
b62d96c Ajout des tests simples dbt
2eb0443 Ajout des modèles dbt
1fac072 Ajout de l'étape de validation minimale
079692a Ajout de l'étape d'ingestion
f3bdd8f Initialisation du projet pipeline
```

III. Schéma simple du pipeline



IV. Réponses aux questions du TP

1. En quoi ce pipeline est-il plus structuré qu'un notebook unique ?

Un notebook unique regroupe tout le code dans un seul endroit, ce qui devient vite désordonné. Ce pipeline est plus structuré car il sépare le code en fichiers spécialisés et modulaires (un script pour l'ingestion, un pour la validation, etc.). Cela rend le projet beaucoup plus facile à tester, à maintenir, et à versionner.

2. Quel est le rôle de Git dans ce projet ?

Le rôle de Git est d'assurer le versionnement du code. Il permet de conserver un historique complet des modifications, de revenir en arrière si un bug apparaît, et de travailler proprement en équipe sans risquer d'écraser le travail des autres.

3. Quel est le rôle de requirements.txt ?

Ce fichier liste précisément toutes les bibliothèques Python nécessaires au projet avec leurs versions. Son rôle est de garantir la reproductibilité de l'environnement : n'importe quel autre utilisateur peut réinstaller le projet à l'identique avec une seule commande (`pip install -r requirements.txt`).

4. Pourquoi ajouter une étape validate.py ?

Cette étape sert de contrôle qualité. Elle vérifie que les données brutes sont conformes (bonnes colonnes, pas de valeurs manquantes critiques) avant d'être transformées. Cela évite qu'une donnée corrompue ne casse les étapes suivantes ou ne fausse les rapports finaux.

5. Quel est le rôle de dbt test ?

Le rôle de dbt test est d'exécuter des tests automatisés directement sur les données transformées dans la base de données. Il permet de s'assurer en continu que les règles métiers sont respectées (par exemple, vérifier qu'une colonne clé n'est jamais nulle).

6. Qu'apporte Dagster par rapport à une exécution manuelle ?

Dagster apporte l'automatisation et la supervision. Au lieu de devoir lancer manuellement chaque script l'un après l'autre dans le terminal, Dagster gère l'ordre d'exécution et les dépendances entre les étapes. Si une étape échoue, il s'arrête proprement et permet d'identifier l'erreur immédiatement via son interface web.

7. Qu'apporte la CI ?

La CI (Intégration Continue) apporte une validation automatique à chaque fois que vous proposez une modification du code. Elle exécute des tests d'intégrité (comme la vérification de la syntaxe Python) pour s'assurer que personne n'envoie de code cassé sur le dépôt principal.

8. Que manque-t-il encore pour un pipeline pleinement industrialisé ?

Pour devenir une solution de production industrielle, il manquerait :

- **Une base de données cloud ou centralisée** : Remplacer DuckDB local par un vrai Data Warehouse (comme PostgreSQL).
- **Une planification temporelle (Scheduling)** : Programmer le pipeline pour qu'il s'exécute automatiquement toutes les heures ou toutes les nuits.
- **Un système d'alertes** : Configurer des notifications automatiques (par Slack ou Email) pour avertir l'équipe en cas de panne ou d'échec du pipeline.
- **La conteneurisation** : Utiliser un outil comme Docker pour packager l'application et la déployer facilement sur un serveur de production.